

Unofficial Nord Electro 5 File Structure Documentation

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Nord Electro 5 File Structures

This part documents the Nord Electro 5 file structures. It was decoded from files written by the instrument and by Nord Sound Manager, changing one setting at a time, and checked on the instrument (OS 2.04). It is not officially supported by Nord Keyboards / Clavia DMI AB. While we certainly hope this document is useful, none of the authors or contributors place any guarantees as to the accuracy of the data.

Nord Electro 5 Summary

- [Nord Electro 5 Program File Structure](#) (ne5p)
- [Nord Electro 5 Live File Structure](#) (ne5l)
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- [Nord Electro 5 Settings File Structure](#) (ne5s)
- [Nord Electro 5 Bundle and Backup File Structure](#) (ne5pbundle, ne5tbundle, ne5b)

Nord Electro 5 Contributors

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Nord Electro 5 Revision

rev	date	description
1.0	27-Sep-2026	Program, live, song, settings, bundle and backup file structures

Nord Electro 5 Program File Structure

This mapping corresponds to the Nord Electro 5 program file (file extension ne5p). A live file (ne5l) has the same body, see [Nord Electro 5 Live File Structure](#).

Offset 0x04 defines the file header format.

type	size	description
0	147	Legacy format: bytes 0x18 to 0x2B are missing, and a CRC-16 is added at the end of the file. The factory programs use it.
1	165	New format with bytes 0x18 to 0x2B (20 bytes), written by Nord Sound Manager 7.40 and later.

The offsets below are for type 1. In a type 0 file the body starts at 0x18, so every body offset is 0x14 lower.

Unlike the Nord Stage program file, the Electro 5 program keeps the complete state of all four organ models and both of their presets at all times, not only the selected one.

Each memory offset corresponds to an 8-bit value. In the documentation below --xxxxxx means bit 5 to bit 0 are used. A value that spans bytes is read most significant bit first: the bits in the earlier byte are the high bits.

offset	bits	description
0x0000	cccccccc	(c) 4-byte Clavia ID, ascii C - 0x43
0x0001	cccccccc	ascii B - 0x42
0x0002	cccccccc	ascii I - 0x49
0x0003	cccccccc	ascii N - 0x4E
0x0004	ffffffff	(f) file format (32-bit, little endian)
0x0005	ffffffff	
0x0006	ffffffff	
0x0007	ffffffff	
0x0008	iiiiiii	(i) 4-byte file ID, ascii n - 0x6E
0x0009	iiiiiii	ascii e - 0x65
0x000A	iiiiiii	ascii 5 - 0x35
0x000B	iiiiiii	ascii p - 0x70
0x000C	bbbbbbbb	(b) bank (16-bit, little endian), 0 = bank 1 . . . 7 = bank 8
0x000D	bbbbbbbb	
0x000E	llllllll	(l) location (16-bit, little endian), 0 = location 1 . . . 49 = location 50
0x000F	llllllll	
0x0010	-----	0xFF
0x0011	-----	0xFF
0x0012	-----	0xFF
0x0013	-----	0xFF
0x0014	vvvvvvvv	(v) version (32-bit, little endian)
0x0015	vvvvvvvv	
0x0016	vvvvvvvv	
0x0017	vvvvvvvv	
0x0018	cccccccc	(c) CRC-32 (32-bit, little endian)
0x0019	cccccccc	
0x001A	cccccccc	
0x001B	cccccccc	
0x001C	-----	0
0x001D	-----	0
0x001E	-----	0
0x001F	-----	0
0x0020	-----	0
0x0021	-----	0
0x0022	-----	0
0x0023	-----	0
0x0024	-----	0
0x0025	-----	0
0x0026	-----	0

offset	bits	description
0x0027	-----	0
0x0028	-----	0
0x0029	-----	0
0x002A	-----	0
0x002B	-----	0
0x002C	vvvvvvvv	(v) version (16-bit, big endian), repeated - start of body
0x002D	vvvvvvvv	
0x002E	llluuuoo	(l) lower part, (u) upper part, (o) lower octave shift
0x002F	oouuuust	(o) lower octave shift, (u) upper octave shift, (s) lower sustain pedal, (t) upper sustain pedal
0x0030	cd-spppt	(c) lower control pedal, (d) upper control pedal, (s) split on, (p) split point, (t) transpose on
0x0031	ttttmmmm	(t) transpose, (m) part mix
0x0032	mmmggggg	(m) part mix, (g) gain
0x0033	ggoooolud	(g) gain, (o) organ model, (l) lower enabled, (u) upper enabled, (d) drawbar live
0x0034	-----	0
0x0035	-----	0
0x0036	-----	0
0x0037	-----	0
0x0038	-----	0
0x0039	-----	0
0x003A	ccc--mmm	(c) piano category, (m) piano model, start of Piano section
0x003B	mm-----c	(m) piano model, (c) clavinet model
0x003C	caattmii	(c) clavinet model, (a) piano acoustics, (t) piano touch, (m) piano mono, (i) piano id (32-bit)
0x003D	iiiiiii	
0x003E	iiiiiii	
0x003F	iiiiiii	
0x0040	iiiiii--	
0x0041	-----	0
0x0042	-----	0
0x0043	-----	0
0x0044	-----	0
0x0045	-----	0
0x0046	aaaaaaad	(a) sample attack, (d) sample decay release, start of Sample section
0x0047	ddddddnn	(d) sample decay release, (n) sample number
0x0048	nnnnnnii	(n) sample number, (i) sample id (32-bit)
0x0049	iiiiiii	
0x004A	iiiiiii	
0x004B	iiiiiii	
0x004C	iiiiidd	(i) sample id (32-bit), (d) sample dynamics (0 to 3)
0x004D	f-----	(f) sample filter
0x004E	-----	start of Organ section
0x004F	-----	
0x0050	-----	
0x0051	vvvhss--	(v) B3 vibrato chorus type, (h) B3 percussion third harmonic, (s) B3 percussion speed
0x0052	-----	
0x0053	-p-----	(p) B3 selected preset
0x0054	-----	unused
0x0055	dddddddd	(d) B3 preset 1 drawbars
0x0056	dddddddd	
0x0057	dddddddd	
0x0058	dddddddd	
0x0059	dddvvpbb	(d) B3 preset 1 drawbars, (v) B3 preset 1 vibrato chorus on, (p) B3 preset 1 percussion on, (b) B3 + Bass bass drawbar 1
0x005A	bbcccc--	(b) B3 + Bass bass drawbar 1, (c) B3 + Bass bass drawbar 2
0x005B	-----	
0x005C	dddddddd	(d) B3 preset 2 drawbars
0x005D	dddddddd	

offset	bits	description
0x005E	dddddddd	
0x005F	dddddddd	
0x0060	dddvp--	(d) B3 preset 2 drawbars, (v) B3 preset 2 vibrato chorus on, (p) B3 preset 2 percussion on, unused
0x0061	-----	unused
0x0062	-----	
0x0063	vvv-----	(v) Vox vibrato chorus type
0x0064	-----	
0x0065	-p-----	(p) Vox selected preset
0x0066	-----	
0x0067	dddddddd	(d) Vox preset 1 drawbars
0x0068	dddddddd	
0x0069	dddddddd	
0x006A	dddddddd	
0x006B	dddv---	(d) Vox preset 1 drawbars, (v) Vox preset 1 vibrato chorus on
0x006C	-----	
0x006D	dddddddd	(d) Vox preset 2 drawbars
0x006E	dddddddd	
0x006F	dddddddd	
0x0070	dddddddd	
0x0071	dddv---	(d) Vox preset 2 drawbars, (v) Vox preset 2 vibrato chorus on
0x0072	-----	
0x0073	vvv-----	(v) Farfisa vibrato chorus type
0x0074	-----	
0x0075	-p-----	(p) Farfisa selected preset
0x0076	-----	
0x0077	dddddddd	(d) Farfisa preset 1 tabs
0x0078	dddddddd	
0x0079	dddddddd	
0x007A	dddddddd	
0x007B	dddv---	(d) Farfisa preset 1 tabs, (v) Farfisa preset 1 vibrato chorus on
0x007C	-----	
0x007D	dddddddd	(d) Farfisa preset 2 tabs
0x007E	dddddddd	
0x007F	dddddddd	
0x0080	dddddddd	
0x0081	dddv---	(d) Farfisa preset 2 tabs, (v) Farfisa preset 2 vibrato chorus on
0x0082	-----	
0x0083	-----	
0x0084	-----	
0x0085	-p-----	(p) Pipe selected preset
0x0086	-----	
0x0087	dddddddd	(d) Pipe preset 1 drawbars
0x0088	dddddddd	
0x0089	dddddddd	
0x008A	dddddddd	
0x008B	ddd----	unused
0x008C	-----	
0x008D	dddddddd	(d) Pipe preset 2 drawbars
0x008E	dddddddd	
0x008F	dddddddd	
0x0090	dddddddd	
0x0091	ddd----	
0x0092	-----	
0x0093	ppttttrr	(p) effect 1 part select, (t) effect 1 type, (r) effect 1 rate, start of Effects section
0x0094	rrrrrppt	(r) effect 1 rate, (p) effect 2 part select, (t) effect 2 type
0x0095	tttrrrrr	(t) effect 2 type, (r) effect 2 rate
0x0096	rrppfftt	(r) effect 2 rate, (p) delay part select, (f) delay feedback (0 to 3), (t) delay tempo
0x0097	tttttwww	(t) delay tempo, (w) delay wet dry

offset	bits	description
0x0098	wwwwpe-f	(w) delay wet dry, (p) delay ping pong, (e) equalizer on, (f) equalizer mid frequency
0x0099	fffffftt	(f) equalizer mid frequency, (t) equalizer treble
0x009A	tttttmmm	(t) equalizer treble, (m) equalizer mid gain
0x009B	mmmmbbbb	(m) equalizer mid gain, (b) equalizer bass
0x009C	bbbppttt	(b) equalizer bass, (p) effect 3 part select, (t) effect 3 type
0x009D	ccccccr	(c) effect 3 compression, (r) reverb on
0x009E	tttwwwww	(t) reverb type, (w) reverb wet dry
0x009F	wwms----	(w) reverb wet dry, (m) rotary speaker stop mode, (s) rotary speaker speed (0 = slow, 1 = fast), unused
0x00A0	-----	unused
0x00A1	---cdee-	unused, (c) effect 1 control pedal, (d) effect 2 deep, (e) equalizer part select
0x00A2	-----	0
0x00A3	-----	0
0x00A4	-----	0

Single bit settings without a link are 0 = off, 1 = on.

NE5 File Format

type	description
0	Legacy header, 24 bytes. Bytes 0x18 to 0x2B are missing, so the body starts at 0x18. The file ends with a 2-byte CRC-16 (IBM-3740, also called CCITT-FALSE, little endian) over every byte before it, header included.
1	Header with checksum, 44 bytes. 0x18 to 0x1B is a CRC-32 (ISO-HDLC / zlib, little endian) over the body only, 0x2C to the end of the file. 0x1C to 0x2B are 0.

The factory programs are type 0. Nord Sound Manager 7.40 and later writes type 1.

NE5 Program Version

Always 4 on every file seen. The body repeats it at 0x2C (16-bit, big endian).

NE5 Part

value	part
0	Organ
1	Piano
2	Sample

NE5 Octave Shift

Stored as the shift plus 7: 1 = -6 . . . 7 = 0 . . . 13 = +6.

NE5 Split Point

value	split point
0	C3
1	F3
2	C4
3	F4
4	C5
5	F5
6	Upper
7	Lower

NE5 Transpose

Transpose is stored as the transposition plus 6: 0 = -6 . . . 6 = 0 . . . 12 = +6.

Transpose on is set the first time the transpose is changed and is never cleared again, so it stays on after the transpose is put back to 0. The transpose light is on only when transpose on is set and the transpose is not 0. While transpose on is clear, transpose holds 7 (+1) and means nothing.

NE5 Part Mix

0 - 127. 64 = both parts at full level. Above 64 the lower part is turned down, below 64 the upper part.

NE5 Knob Scale

Gain, delay wet dry, reverb wet dry and effect 3 compression are 0 - 127 on a linear scale, shown as 0 - 10 on the panel.

NE5 Organ Model

value	organ model
0	B3
1	B3 + Bass
2	Pipe
3	Vox
4	Farfisa

NE5 Drawbar Live

0 = off, 1 = on. Drawbar Sync is a momentary command and is not stored.

NE5 Piano Category

value	category
0	Grand
1	Upright
2	Electric Piano 1
3	Electric Piano 2
4	Clavinet
5	Harpsichord

NE5 Piano Model

0 = first model in the category, 1 = second . . .

Category and model together are the piano's position in the instrument's library. The library is organised one bank per category, so category is the bank and model is the slot within it, both counted from 0: the piano in bank 3 slot 2 stores 2/1. These are positions, so they move if the library is reorganised - the piano id is the stable reference.

NE5 Clavinet Model

0 to 3. Only set when the piano category is Clavinet, 0 otherwise.

NE5 Piano Acoustics

value	acoustics
0	Off
1	String Resonance
2	Long Release
3	String Resonance and Long Release

value	acoustics
-------	-----------

The names come from the settings the test files were made with. String Resonance was confirmed by ear; Long Release made no audible difference on the piano it was tried with.

NE5 Piano Touch

0 = off, 1 = touch 1, 2 = touch 2, 3 = touch 3.

Touch remaps the velocity played rather than the level. All three settings lift soft notes by about 10 velocity steps; the higher the setting, the more the middle of the range is lifted too.

NE5 Piano and Sample ID

32-bit identifier of the piano (npno) or sample (nsmp) the program depends on, 0 = none.

This is the identity of the piano or sample, independent of where it sits in the library. The instrument reports the same value when asked for a program's dependencies over USB, which is how a program can be matched to what it needs.

NE5 Sample Attack

0 - 127, 0 = 0.5 ms to 127 = 45 s. The curve in between is not known.

NE5 Sample Decay Release

One knob covers both: 0 = 3 ms decay . . . 64 = sustain . . . 127 = longest release. Below 64 it sets a decay, above 64 a release, and 64 (sustain) is the knob's centre detent.

NE5 Sample Number

Position of the sample in the Samp Lib, counted from 0 (0 = the sample shown as 1 on the panel).

This is a panel position only. It does not match the sample's slot in the instrument's sample library, and it is reused - the same number appears against different samples, and the same sample appears under different numbers. The instrument does not use it to pick the sample: a program whose number and id disagree plays the sample the id names. Use the sample id to identify a sample.

NE5 Organ Preset

0 = preset 1, 1 = preset 2. One per organ model.

NE5 Drawbars

9 drawbars per model and preset, packed one per nibble, high nibble first. Each nibble is the physical drawbar position, 0 to 8, stored as its own value.

Farfisa has on/off tabs rather than continuous drawbars. The file still stores a 0 to 8 position per tab, and 5 or higher reads as on, below 5 as off. The exact position below or above the threshold varies between files that show the same tab state.

In B3 + Bass, preset 1's first two nibbles are not used, see [NE5 B3 Bass Drawbars](#).

NE5 Vibrato Chorus Type

One per model, shared by both presets. Each model offers a different set of modes, so the same stored value means different things depending on the model. Pipe has no vibrato chorus.

value	B3	Vox	Farfisa
0	V1	V1	V1
1	C1	V2	V2
2	V2	V3	C2
3	C2		C3

value	B3	Vox	Farfisa
4	V3		
5	C3		

NE5 B3 Percussion

Percussion on is per preset (0x59 b2 for preset 1, 0x60 b2 for preset 2). Third harmonic and speed are shared.

Third harmonic: 0 = off (second harmonic), 1 = on (third harmonic).

value	speed
0	Off
1	Fast
2	Soft
3	Both

The stored speed values are not in panel order.

NE5 B3 Bass Drawbars

When the organ model is B3 + Bass, the two presets are different instruments. Preset 1 is the bass manual and only has the two bass drawbars; preset 2 is an ordinary B3 and uses the normal nine nibbles at 0x5C.

The bass registration is not in preset 1's nine-nibble block at 0x55, which holds stale values in this mode: moving the bass drawbars leaves its first two nibbles unchanged, neither 0 nor the bass values. The two bass drawbars are 0 to 8, stored as themselves:

```
bass drawbar 1 = ((byte at 0x59 AND 0x03) << 2) OR (byte at 0x5A >> 6)
bass drawbar 2 = (byte at 0x5A >> 2) AND 0x0F
```

0x59 b3 and b2 are the B3 preset 1 vibrato chorus and percussion bits, so mask them off. Whether the bass manual responds to them is not known.

NE5 Organ Unused Bits

- 0x54: always 0. The keyboard writes 0 here every time a program is stored.
- 0x60 b1-0 and 0x61 b7-2: these sit where a preset 2 copy of the B3 + Bass bass drawbars would be, with the same layout, and fresh programs hold 8 and 8 in them (0x60 b1-0 = 2, 0x61 = 0x20). The keyboard (firmware 2.04) neither reads nor writes them: other values put here change nothing, and storing the program keeps them.
- 0x8B b3: where the other models keep their preset 1 vibrato chorus on. It is 1 in almost every file, but the vibrato chorus button does not respond while Pipe is selected, so it cannot be changed from the panel.

NE5 Effect Part Select

Effects 1, 2, 3 and the delay each have a part select that doubles as their on/off.

value	part
0	Off
1	Off, as written by older firmware
2	Lower
3	Upper

The keyboard treats a stored 1 as off and keeps it when the program is stored again. Current firmware writes 0.

NE5 Effect 1 Type

value	type
0	TREM1
1	TREM2
2	TREM1&2
3	PAN1
4	PAN2
5	PAN1&2
6	WAH
7	RM

The stored values are not in panel order.

NE5 Effect 1 Rate

0 - 127 on a linear scale, shown as 1 - 10 on the panel.

NE5 Effect 2 Type

value	type
0	PHAS1
1	PHAS2
2	FLANGE
3	CHOR1
4	CHOR2
5	VIBE

The stored values are not in panel order.

NE5 Effect 2 Rate

0 - 127, 0 = 0 Hz to 127 = 10.5 Hz. The curve in between is not known.

NE5 Effect 3 Type

value	type
0	None
1	Small
2	JC
3	Twin
4	Rotary
5	Comp

The stored values are not in panel order.

NE5 Delay Tempo

0 - 127, running backwards: 0 = 750 ms and 127 = 20 ms.

NE5 Reverb Type

value	type
0	Room
1	Stage Soft
2	Stage
3	Hall Soft
4	Hall

value	type
-------	------

The stored values are not in panel order.

NE5 Equalizer

Equalizer on (0x98 b2) and the equalizer part select (0xA1 b2-1) are in different bytes and have to be read together: part select 0 means Lower, not off.

value	part
0	Lower
1	Upper
2	Lower + Upper

Bass, mid gain and treble: 0 - 127, 0 = -15 dB, 64 = 0 dB (centre), 127 = +15 dB.

Mid frequency: 0 - 127, 0 = 200 Hz to 127 = 8 kHz. The curve in between is not known.

NE5 Dead Trailing Bits

0x9F b2-0, 0xA0 and 0xA1 b7-5 are not read by the keyboard. Usually 0, but some older files carry other values here. They make no difference to the sound or the panel, and storing the program on the keyboard keeps them unchanged.

Nord Electro 5 Live File Structure

This mapping corresponds to the Nord Electro 5 live file (file extension ne5l). The Electro 5 has three Live slots, Live 1 to Live 3, and a live file holds one of them. Nord Sound Manager exports them as Live 1.ne5l to Live 3.ne5l, and a full backup (ne5b) carries them as Live/Live/Live 1.ne5l and so on.

The body is the program body. A live file and a program file (ne5p) have the same size and the same 121-byte body at 0x2C to 0xA4, byte for byte, so every body offset in the [Nord Electro 5 Program File Structure](#) applies unchanged. The same panel state read from the instrument once as Live 1 and once as a program gave identical bodies.

Only the header differs: the file ID is ne5l, and the location addresses the three Live slots, not the 8 banks of 50 programs. A live file has no program bank or program location.

offset	bits	description
0x0000	cccccccc	(c) 4-byte Clavia ID, ascii C - 0x43
0x0001	cccccccc	ascii B - 0x42
0x0002	cccccccc	ascii I - 0x49
0x0003	cccccccc	ascii N - 0x4E
0x0004	ffffffff	(f) file format (32-bit, little endian)
0x0005	ffffffff	
0x0006	ffffffff	
0x0007	ffffffff	
0x0008	iiiiiii	(i) 4-byte file ID, ascii n - 0x6E
0x0009	iiiiiii	ascii e - 0x65
0x000A	iiiiiii	ascii 5 - 0x35
0x000B	iiiiiii	ascii l - 0x6C
0x000C	bbbbbbbb	(b) bank (16-bit, little endian), always 0
0x000D	bbbbbbbb	
0x000E	11111111	(l) location (16-bit, little endian), 0 = Live 1, 1 = Live 2, 2 = Live 3
0x000F	11111111	
0x0010	-----	0xFF
0x0011	-----	0xFF
0x0012	-----	0xFF
0x0013	-----	0xFF
0x0014	vvvvvvvv	(v) version (32-bit, little endian)
0x0015	vvvvvvvv	
0x0016	vvvvvvvv	
0x0017	vvvvvvvv	
0x0018	cccccccc	(c) CRC-32 (32-bit, little endian)
0x0019	cccccccc	
0x001A	cccccccc	
0x001B	cccccccc	
0x001C	-----	0
0x001D	-----	0
0x001E	-----	0
0x001F	-----	0
0x0020	-----	0
0x0021	-----	0
0x0022	-----	0
0x0023	-----	0
0x0024	-----	0
0x0025	-----	0
0x0026	-----	0
0x0027	-----	0
0x0028	-----	0
0x0029	-----	0
0x002A	-----	0
0x002B	-----	0
0x002C	vvvvvvvv	(v) version (16-bit, big endian), repeated - start of body
0x002D	vvvvvvvv	0x002E to 0x00A4: same as the program body

NE5 Live File Format

The same two header types as the program file: type 0 (147 bytes, no 0x18 to 0x2B, body at 0x18, CRC-16 at the end of the file over every byte before it) and type 1 (165 bytes, CRC-32 at 0x18 over the body only). The factory live files are type 0.

Because the type 1 CRC-32 only covers the body, a type 1 live file and a type 1 program file with the same body and the same location word differ in one byte only, 0x0B (l against p). To turn one into the other, change the file ID and set the location for the new slot space; the checksum at 0x18 stays valid. In a type 0 file the CRC-16 covers the header as well, so it must be recomputed after any header change.

NE5 Live Version

Always 4 on every file seen, the same as a program. The body repeats it at 0x2C (16-bit, big endian).

Nord Electro 5 Song File Structure

This mapping corresponds to the Nord Electro 5 song file (file extension ne5t). A song is a set of four program slots, A to D. The Electro 5 holds 4 banks of 50 songs.

Offset 0x04 defines the file header format.

type	size	description
0	44	Legacy format: bytes 0x18 to 0x2B are missing, and a CRC-16 is added at the end of the file. The factory songs use it.
1	62	New format with bytes 0x18 to 0x2B (20 bytes), written by Nord Sound Manager 7.40 and later.

The offsets below are for type 1. In a type 0 file the body starts at 0x18, so every body offset is 0x14 lower.

The song name is not stored in the file. The instrument keeps it separately (16 bytes) and sends it next to the body over USB.

offset	bits	description
0x0000	cccccccc	(c) 4-byte Clavia ID, ascii C - 0x43
0x0001	cccccccc	ascii B - 0x42
0x0002	cccccccc	ascii I - 0x49
0x0003	cccccccc	ascii N - 0x4E
0x0004	ffffffff	(f) file format (32-bit, little endian)
0x0005	ffffffff	
0x0006	ffffffff	
0x0007	ffffffff	
0x0008	iiiiiii	(i) 4-byte file ID, ascii n - 0x6E
0x0009	iiiiiii	ascii e - 0x65
0x000A	iiiiiii	ascii 5 - 0x35
0x000B	iiiiiii	ascii t - 0x74
0x000C	bbbbbbbb	(b) bank (16-bit, little endian), 0 = bank 1 . . . 3 = bank 4
0x000D	bbbbbbbb	
0x000E	llllllll	(l) location (16-bit, little endian), 0 = location 1 . . . 49 = location 50
0x000F	llllllll	
0x0010	-----	0xFF
0x0011	-----	0xFF
0x0012	-----	0xFF
0x0013	-----	0xFF
0x0014	vvvvvvvv	(v) version (32-bit, little endian)
0x0015	vvvvvvvv	
0x0016	vvvvvvvv	
0x0017	vvvvvvvv	
0x0018	cccccccc	(c) CRC-32 (32-bit, little endian)
0x0019	cccccccc	
0x001A	cccccccc	
0x001B	cccccccc	
0x001C	-----	0
0x001D	-----	0
0x001E	-----	0
0x001F	-----	0
0x0020	-----	0
0x0021	-----	0
0x0022	-----	0
0x0023	-----	0
0x0024	-----	0
0x0025	-----	0
0x0026	-----	0
0x0027	-----	0
0x0028	-----	0

offset	bits	description
0x0029	-----	0
0x002A	-----	0
0x002B	-----	0
0x002C	vvvvvvvv	(v) version (16-bit, big endian), repeated - start of body
0x002D	vvvvvvvv	
0x002E	aaaaaaaa	(a) program slot A
0x002F	bbbbbbbb	(a) program slot A, (b) program slot B
0x0030	bbcccccc	(b) program slot B, (c) program slot C
0x0031	cccdcccc	(c) program slot C, (d) program slot D
0x0032	dddd----	
0x0033	-----	0
0x0034	-----	0
0x0035	-----	0
0x0036	-----	0
0x0037	-----	0
0x0038	-----	0
0x0039	-----	0
0x003A	-----	0
0x003B	-----	0
0x003C	-----	0
0x003D	-----	0

NE5 Song File Format

type	description
0	Legacy header, 24 bytes. Bytes 0x18 to 0x2B are missing, so the body starts at 0x18. The file ends with a 2-byte CRC-16 (IBM-3740, also called CCITT-FALSE, little endian) at 0x2A over every byte before it, header included.
1	Header with checksum, 44 bytes. 0x18 to 0x1B is a CRC-32 (ISO-HDLC / zlib, little endian) over the body only, 0x2C to 0x3D. 0x1C to 0x2B are 0.

NE5 Song Version

value	version
0	the songs shipped on the instrument from the factory
1	any song the instrument has written, by the user or as a side effect

The instrument never writes version 0. A factory song becomes version 1 as soon as the instrument rewrites it, even when nothing asked for that: moving a program rewrites every song that uses it.

The body repeats the version at 0x2C (16-bit, big endian). Only 0 and 1 have been seen, so in practice this is a single bit, 0x2D b0. Apart from the header field at 0x14 (and so the checksum), this is the only difference between a version 0 and a version 1 song: the two copies always agree, and nothing else in the body changes with the version. The body is also the only part the instrument sends when a song is read from it - the header is not transmitted - so this looks like the version being carried where the receiving side can see it.

NE5 Song Program Slot

Each slot is a 9-bit program location, counted straight through the program banks:

$$\text{value} = (\text{program bank} - 1) * 50 + (\text{program location} - 1)$$

so 0 = program 1:1, 49 = program 1:50, 50 = program 2:1 . . . 399 = program 8:50. 400 to 511 are never seen.

Moving a program on the instrument rewrites the slots of every song that uses it, so they follow the program. Deleting a program does not: the slots keep pointing at the now empty location.

Nord Electro 5 Settings File Structure

This mapping corresponds to the Nord Electro 5 system settings file (file extension ne5s). The instrument holds exactly one. It carries the System, MIDI and Sound menus, plus the selection the instrument restores at power-up. Nord Sound Manager exports it as Settings.ne5s, and a full backup (ne5b) carries it as Settings/Settings/Settings.ne5s.

Offset 0x04 defines the file header format.

type	size	description
0	60	Legacy format: bytes 0x18 to 0x2B are missing, and a CRC-16 is added at the end of the file. The factory settings use it.
1	78	New format with bytes 0x18 to 0x2B (20 bytes), written by Nord Sound Manager 7.40 and later.

The offsets below are for type 1. In a type 0 file the body starts at 0x18, so every body offset is 0x14 lower.

Every setting in the System, MIDI and Sound menus is in this file except two: Memory Protect (System) and Local Control (MIDI) are not stored here at all. Toggling either one on the panel and reading the settings back changes no bit of the file, so they are absent, not undecoded. Each setting below was found by changing that one setting on the panel and reading the file back; the change moves exactly the bits listed and nothing else.

offset	bits	description
0x0000	cccccccc	(c) 4-byte Clavia ID, ascii C - 0x43
0x0001	cccccccc	ascii B - 0x42
0x0002	cccccccc	ascii I - 0x49
0x0003	cccccccc	ascii N - 0x4E
0x0004	ffffffff	(f) file format (32-bit, little endian)
0x0005	ffffffff	
0x0006	ffffffff	
0x0007	ffffffff	
0x0008	iiiiiii	(i) 4-byte file ID, ascii n - 0x6E
0x0009	iiiiiii	ascii e - 0x65
0x000A	iiiiiii	ascii 5 - 0x35
0x000B	iiiiiii	ascii s - 0x73
0x000C	bbbbbbbb	(b) bank (16-bit, little endian), always 0
0x000D	bbbbbbbb	
0x000E	llllllll	(l) location (16-bit, little endian), always 0
0x000F	llllllll	
0x0010	-----	0xFF
0x0011	-----	0xFF
0x0012	-----	0xFF
0x0013	-----	0xFF
0x0014	vvvvvvvv	(v) version (32-bit, little endian)
0x0015	vvvvvvvv	
0x0016	vvvvvvvv	
0x0017	vvvvvvvv	
0x0018	cccccccc	(c) CRC-32 (32-bit, little endian)
0x0019	cccccccc	
0x001A	cccccccc	
0x001B	cccccccc	
0x001C	-----	0
0x001D	-----	0
0x001E	-----	0
0x001F	-----	0
0x0020	-----	0
0x0021	-----	0
0x0022	-----	0
0x0023	-----	0
0x0024	-----	0

offset	bits	description
0x0025	-----	0
0x0026	-----	0
0x0027	-----	0
0x0028	-----	0
0x0029	-----	0
0x002A	-----	0
0x002B	-----	0
0x002C	vvvvvvvv	(v) version (16-bit, big endian), repeated - start of body
0x002D	vvvvvvvv	
0x002E	sl-nnppp	(s) set list mode, (l) live mode, (n) live slot, (p) program
0x002F	ppppppss	(p) program, (s) song
0x0030	sssssscc	(s) song, (c) MIDI control change mode
0x0031	pp-ssccc	(p) MIDI program change mode, (s) sustain pedal type, (c) control pedal type
0x0032	----rrmf	(r) rotary pedal type, (m) rotary pedal mode, (f) fine tune
0x0033	ffffff--	
0x0034	rrrrtttt	(r) piano string resonance, (t) global transpose
0x0035	ggggg--r	(g) MIDI global channel, (r) rotary speaker type
0x0036	rrhhhsss	(r) rotary speaker type, (h) rotary horn speed, (s) rotary rotor speed
0x0037	hhhaaabb	(h) rotary horn acceleration, (a) rotary rotor acceleration, (b) rotary balance
0x0038	bffsssn	(b) rotary balance, (f) B3 percussion decay fast, (s) B3 percussion decay slow, (n) B3 percussion volume normal
0x0039	nnsssttt	(n) B3 percussion volume normal, (s) B3 percussion volume soft, (t) B3 tonewheel mode
0x003A	mkk-btll	(m) B3 percussion drawbar 9 mute, (k) B3 key click level, (b) B3 key bounce, (t) B3 trig mode (0 = normal, 1 = fast), (l) MIDI lower part receive channel
0x003B	llluuuuu	(l) MIDI lower part receive channel, (u) MIDI upper part receive channel
0x003C	osssssmm	(o) output routing (0 = stereo, 1 = lower left / upper right), (s) MIDI upper split channel, (m) sustain pedal mode
0x003D	-tgggg--	(t) MIDI transpose at (0 = MIDI in, 1 = MIDI out), (g) control pedal gain
0x003E	-----	0
0x003F	-----	0
0x0040	-----	0
0x0041	-----	0
0x0042	-----	0
0x0043	-----	0
0x0044	-----	0
0x0045	-----	0
0x0046	-----	0
0x0047	-----	0
0x0048	-----	0
0x0049	-----	0
0x004A	-----	0
0x004B	-----	0
0x004C	-----	0
0x004D	-----	0

Single bit settings without a link are 0 = off, 1 = on.

NE5 Settings File Format

type	description
0	Legacy header, 24 bytes. Bytes 0x18 to 0x2B are missing, so the body starts at 0x18. The file ends with a 2-byte CRC-16 (IBM-3740, also called CCITT-FALSE, little endian) over every byte before it, header included.
1	Header with checksum, 44 bytes. 0x18 to 0x1B is a CRC-32 (ISO-HDLC / zlib, little endian) over the body only, 0x2C to 0x4D. 0x1C to 0x2B are 0.

NE5 Settings Version

Always 0 on every file seen. The body repeats it at 0x2C (16-bit, big endian).

NE5 Settings Power Up Selection

These are not menu settings. They are where the instrument was, and what it returns to when switched on. Each keeps the last selection of its own mode, so for example the live slot is kept while live mode is off.

Set list mode and live mode: 0 = off, 1 = on. Set list mode is inferred from the files, not confirmed at the panel: the one file taken in set list mode has it set, while other files have a different song selected with it clear, so it follows the mode rather than the song.

Live slot: 0 = Live 1, 1 = Live 2, 2 = Live 3. 3 is never seen.

Program is a 9-bit program location, counted straight through the program banks, the same way the song file counts them:

$\text{value} = (\text{program bank} - 1) * 50 + (\text{program location} - 1)$

so 0 = program 1:1, 49 = program 1:50, 50 = program 2:1 . . . 399 = program 8:50.

Song is an 8-bit song location, counted the same way over the 4 song banks: 0 = song 1:1 . . . 199 = song 4:50. Inferred from the files, not confirmed at the panel.

NE5 Settings MIDI Message Mode

Control change mode and program change mode.

value	mode
0	Off
1	Send
2	Receive
3	Send & Receive

NE5 Settings MIDI Channel

Global channel, lower and upper part receive channel, and upper split channel.

0 = channel 1, 1 = channel 2 . . . 15 = channel 16, 16 = off.

NE5 Settings Pedals

value	sustain pedal type	control pedal type	rotary pedal type
0	Auto	Roland EV-7	Closed
1	Closed	Yamaha FC-7	Open
2	Open	Korg EXP-2	Half Moon
3		Korg XVP-10	
4		Boss FV-500L	
5		Fatar SL	

Rotary pedal mode: 0 = Hold, 1 = Toggle.

Sustain pedal mode: 0 = Sustain, 1 = Sustain + Rotor Hold, 2 = Sustain + Rotor Toggle.

Control pedal gain is stored one less than the panel shows: 0 = 1, 1 = 2 . . . 9 = 10. 1 and 10 were captured; 2 to 9 are assumed to follow the same scale.

NE5 Settings Fine Tune

0 = -50 cents . . . 50 = 0 . . . 100 = +50 cents. -50, 0, +5 and +50 were captured, and -50, 0 and +5 were written back to the instrument and measured as that many cents of pitch change. The values in between follow the same scale but were not each captured.

NE5 Settings Transpose and Resonance

Global transpose: 0 = -6 . . . 6 = 0 . . . 12 = +6 semitones.

Piano string resonance: 0 = -6 dB . . . 6 = 0 dB . . . 12 = +6 dB. Only -6, 0 and +6 dB were captured; the values in between are assumed to follow the same encoding as global transpose.

NE5 Settings Rotary Speaker

Rotary speaker type: 0 = 122, 1 = 122 Close. The panel offers more types; their values were not captured.

Horn speed, rotor speed, horn acceleration and rotor acceleration: 0 = Low, 1 = Normal, 2 = High.

value	balance (bass / horn)
0	70/30
1	60/40
2	50/50
3	40/60
4	30/70

NE5 Settings B3 Organ

Percussion decay fast and slow: 0 = Short, 1 = Medium, 2 = Long.

Percussion volume normal and soft: 0 = Low, 1 = Medium, 2 = High.

Tonewheel mode: 0 = Clean, 1 = Vintage 1, 2 = Vintage 2, 3 = Vintage 3.

Key click level: 0 = Low, 1 = Normal, 2 = High, 3 = Higher.

Nord Electro 5 Bundle and Backup File Structure

This mapping corresponds to the Nord Electro 5 bundle and backup files.

extension	description
ne5pbundle	program bundle: the chosen programs plus every piano and sample they use
ne5tbundle	song (set list) bundle: the song, the programs it uses, and every piano and sample those programs use
ne5b	full backup: settings, live slots, programs, songs, pianos and samples

All three are ordinary ZIP archives. They are not CBIN files and have no Nord header of their own. Every member is a normal Nord file, byte for byte the same as the standalone file (ne5p, ne5t, ne5s, ne5l, npno, nsmp), so the other Electro 5 structures apply to the members unchanged. The only extra member is a small XML manifest called meta.xml.

This layout is not Electro 5 specific. Backups of other Nord models seen so far are the same stored ZIP with a meta.xml, so the extension (or the member file extensions) is what identifies the model, not meta.xml.

NE5 Bundle ZIP Layer

Every member is stored (ZIP compression method 0), never deflated, so each member's bytes follow its local file header unchanged.

Member paths are <folder>/<sub folder>/<name>.<ext>, for example:

```
Program/Bank 7/<program name>.ne5p
Set List/Set List 1/<song name>.ne5t
Piano/Grand/<piano name>.npno
Samp Lib/Samp Lib/<sample name>.nsmp
Settings/Settings/Settings.ne5s
Live/Live/Live 1.ne5l
meta.xml
```

For programs and songs the file name is the program or song name, which the file itself does not store anywhere. The folder names the bank; the location inside the bank is only in the member's own header (0x0C to 0x0F). For pianos the sub folder is the piano category.

Every central directory entry carries a 1-byte extra field, ascii "0" (0x30) or "1" (0x31). The local file headers carry no extra field. Its meaning is unknown. It is not alignment padding (no link to name length or offset) and not "shared by several programs"; both values appear across every kind of member. The ZIP specification asks for at least 4 bytes in an extra field, so this 1-byte field is a Nord Sound Manager quirk.

NE5 Bundle Manifest

meta.xml in a bundle (ne5pbundle, ne5tbundle) has a <bundle> root element, then one <file> element per program or song:

```
<bundle version="1" product="39" product_version="204"
  content_version="1" source="-1">
  <file name="Program/Bank 7/<program name>.ne5p" depCnt="2"
    dep0="Piano/Harps/<piano name>.npno"
    dep1="Samp Lib/Samp Lib/<sample name>.nsmp"/>
  . . .
</bundle>
```

attribute	description
name	member path of the program or song
depCnt	number of dependencies that follow
dep0 . . .	member path of each dependency, in the same archive: for a song its programs (up to 4), for a program its piano and/or sample

On every Electro 5 bundle seen so far the root attributes are version="1", product="39", content_version="1" and source="-1". product="39" is the same number as product_id in an Electro 5 backup. product_version="204"

matches the instrument's OS version (2.04) when those bundles were made. The meaning of version, content_version and source is unknown.

NE5 Backup Manifest

meta.xml in a backup (ne5b) is a single empty <backup> element with attributes only and no file list (a backup holds everything, so no dependency list is needed):

```
<backup file_version="1" backup_format_version="1" product_id="39"
        product_version="204" product_build="592"
        product_content_version="1" manager_version="920"
        manager_build="1462"/>
```

attribute	description
product_id	39 on every Electro 5 backup seen. Backups of other models carry other numbers, so this identifies the model.
product_version	instrument OS version x 100: 204 = OS 2.04, 144 on the factory backup whose content is OS 1.44
manager_version	Nord Sound Manager version x 100 (920 = 9.20)
product_build	build number that goes with product_version (592 with 204, 532 with 144)
manager_build	build number that goes with manager_version (1462 with 920, 890 with 731)
file_version, backup_format_version, product_content_version	always 1 on Electro 5 backups; meaning unknown